Data Analytics Mini Project Guidelines

Project Objective

The mini project aims to assess your ability to perform real-world data analysis using **Excel** for data cleaning and transformation and **Power BI** for data visualization and report building.

This mini project carries a total of 25 marks.

Project Scope

Your project will involve the following key components:

1. **Data Pre-processing using Excel**
    - Perform cleaning and transformation tasks on raw data, such as:
        - Removing duplicates
        - Handling missing values
        - Standardizing formats
        - Creating calculated fields (if required)
        - Filtering and sorting
        - Creating Pivot Tables (if required)

2. **Data Visualization using Power BI**
    - Build a report/dashboard with:
        - Multiple visualizations that communicate insights clearly
        - Appropriate use of charts (bar, line, pie, slicers, cards, etc.)
        - Clear and interactive layout (using filters, slicers, drill-downs, etc.)
        - Proper titles, labels, and legends

Project Domains

Here are some sample domain ideas to help you get started. However, you are free to choose any domain or topic of your interest.

1. **Sales & E-commerce**
    - Sales performance dashboard
    - Customer segmentation based on region or product category
    - Trend and seasonality analysis of sales data

2. **Finance**
    - Expense tracking dashboard
    - Budget vs. Actual analysis
    - Profitability by product or region

3. **Healthcare**
    - Hospital/clinic patient records analysis
    - Disease count trends across months/years
    - Treatment cost analysis

4. **Retail**
    - Inventory and stock analysis
    - Customer purchasing patterns
    - Store-wise performance comparison

5. **Education**
    - Student performance dashboard
    - Course enrollment and completion analysis
    - Feedback and satisfaction trends

6. **Human Resources (HR)**
    - Employee attrition or retention reports
    - Department-wise performance or salary comparison
    - Leave and attendance trends

7. **Environmental**
    - Traffic pattern analysis across regions or time periods
    - Air quality monitoring and pollutant trends
    - Water consumption or waste generation tracking

8. **Social Media & Public Sentiment**
    - Sentiment analysis of social media posts or movie reviews (e.g., Twitter, IMDB)
    - Trending hashtags or topics analysis
    - Public opinion analysis on a campaign or event

Dataset Requirements

The dataset should have:

- ➢ **Minimum of 500 records**
- ➢ **At least 10 features/columns** (preferably a mix of numerical and categorical data)

Data Sources

Here are some reliable sources where you can find datasets for your project:

1. **UCI Machine Learning Repository**: UCI Repository
2. **Google Dataset Search**: Google Dataset Search
3. **Data.gov (U.S.)**: Data.gov
4. **Data.gov.in (India)**: Data.gov.in
5. **World Health Organization (WHO)**: WHO Data
6. **India Data Portal**: India Data Portal
7. **Awesome Public Datasets**: Awesome Public Datasets
8. **OpenML**: OpenML
9. **FiveThirtyEight**: FiveThirtyEight Datasets
10. **Hugging Face** https://huggingface.co/datasets

Submission Requirements

You must submit the following files:

1. **Original Dataset Source**
   ➢ Provide the dataset link from where the raw data was sourced.

2. **Cleaned Dataset**
   ➢ Submit the .xlsx Excel file after performing data cleaning and transformation.

3. **Power BI Report (PDF Export)**
   ➢ Export and submit your Power BI report/dashboard as a **PDF file** (.pdf).

Mini-Project Rubrics [Total: 25 marks]

TASK 1: Dataset Selection (5 marks)

- Relevance, size, and source of the dataset.

TASK 2: Data Pre-processing using Excel (10 marks)

- Cleaning, transformation, and Excel features were used.

TASK 3: Data Visualization using Power BI (10marks)

- Visualizations, insights, and design of the report/dashboard.
